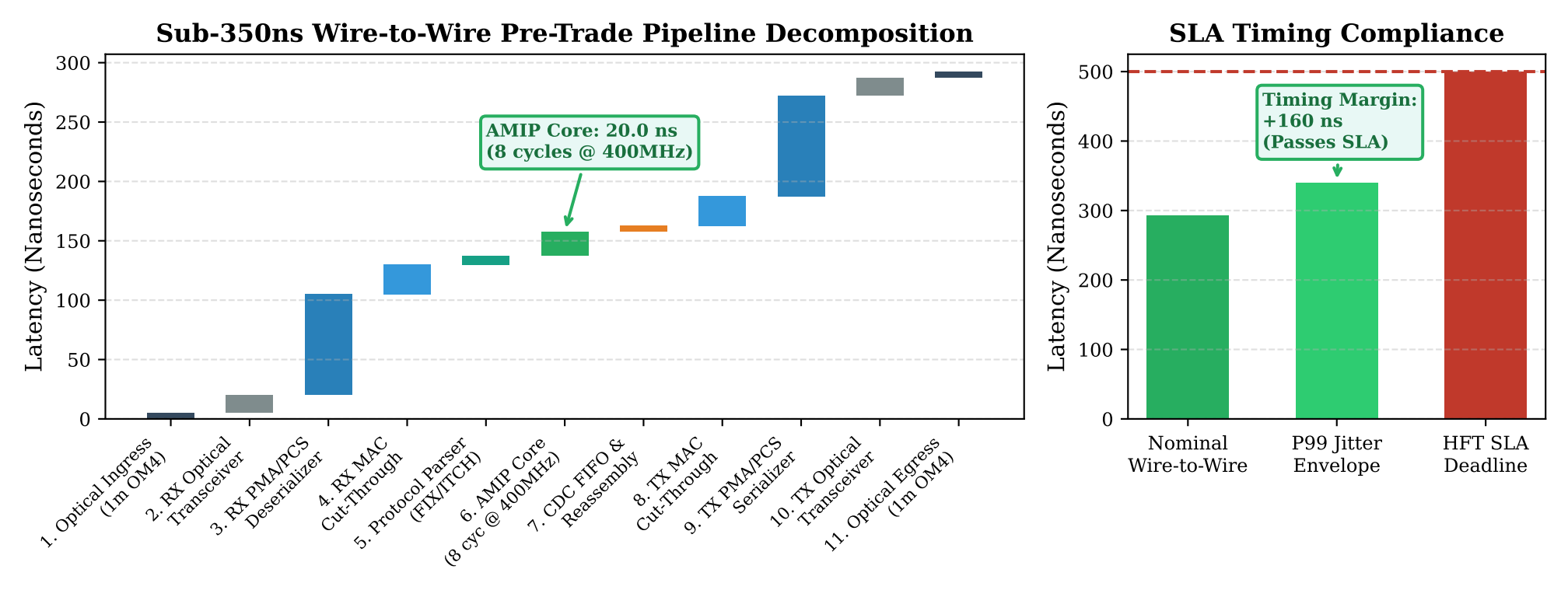
Sub-350ns Wire-to-Wire Pre-Trade Pipeline Decomposition
Latency (Nanoseconds)
0
50
100
150
200
250
300
AMIP Core: 20.0 ns
(8 cycles @ 400MHz)
1. Optical Ingress (1m OM4)
2. RX Optical Transceiver
3. RX PMA/PCS Deserializer
4. RX MAC Cut-Through
5. Protocol Parser (FIX/ITCH)
6. AMIP Core (8 cyc @ 400MHz)
7. CDC FIFO & Reassembly
8. TX MAC Cut-Through
9. TX PMA/PCS Serializer
10. TX Optical Transceiver
11. Optical Egress (1m OM4)
SLA Timing Compliance
Latency (Nanoseconds)
0
100
200
300
400
500
Timing Margin: +160 ns (Passes SLA)
Nominal Wire-to-Wire
P99 Jitter Envelope
HFT SLA Deadline